

Leonardo Pelorosso

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PROJECTS

montecarlo-DGL

- Engineered a high-performance C++ library for N-dimensional Monte Carlo integration (up to 12D), featuring Importance Sampling, Metropolis-Hastings MCMC, runtime expression parsing via muParserX, and convex polytope integration using Qhull.
- Developed thread-safe parallel stochastic optimizers (Particle Swarm Optimization and Genetic Algorithms) utilizing OpenMP, achieving up to 10x speedups on multi-core architectures.
- Architected a geometric optimization solver for drone arm balancing, combining PSO for rapid exploration with a 1M-sample Monte Carlo verification to achieve a center of mass precision error below 0.1%.
- Designed a 15-turbine wind farm layout simulator to maximize power generation under Weibull distributions and wake effects, benchmarking PSO against GA coupled with Metropolis-Hastings integration for power estimation.

EDUCATION

Politecnico di Milano

2024 – Present

Master in High Performance Computing

Milano, MI

- **Coursework:** Computer Architecture, Parallel Computing, Advanced Methods for Scientific Computing, Applied Statistics, Artificial Neural Networks and Deep Learning, Quantum Computing, Quantum Communications, Physics of Nuclear Materials & Nuclear Technologies for Analysis of Materials

Politecnico di Milano

2021 – 2024

Bachelor in Engineering Physics

Milano, MI

- **Coursework:** Principles of Physics, Electromagnetism and Optics, Fundamentals of Quantum Physics, Structure of the Matter, Introduction to Nuclear Physics, Principles of Lasers, Principles of Electronics, Mathematical Analysis (I, II, III)

SKILLS

Programming Languages : C/C++, Python

DB Technologies :SQL (PostgreSQL), MongoDB, Neo4j (Cypher), Elasticsearch, Cassandra, REDIS

HPC & Parallel : CUDA, MPI, OpenMP, GPU Computing, Cluster environments

Systems : MacOS, Linux (Ubuntu), Windows

Cooperative Tool : Git, Docker, Slurm

Spoken Languages Italian (Native), English(C1 - CAE:190), Spanish (B1)